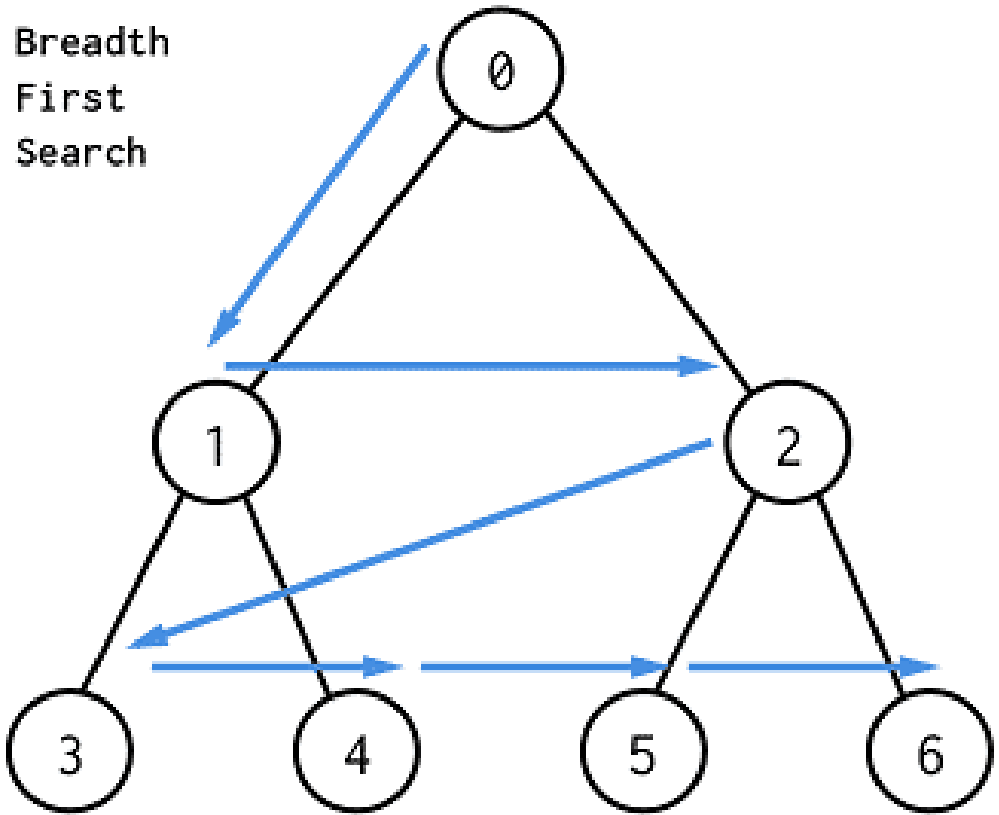
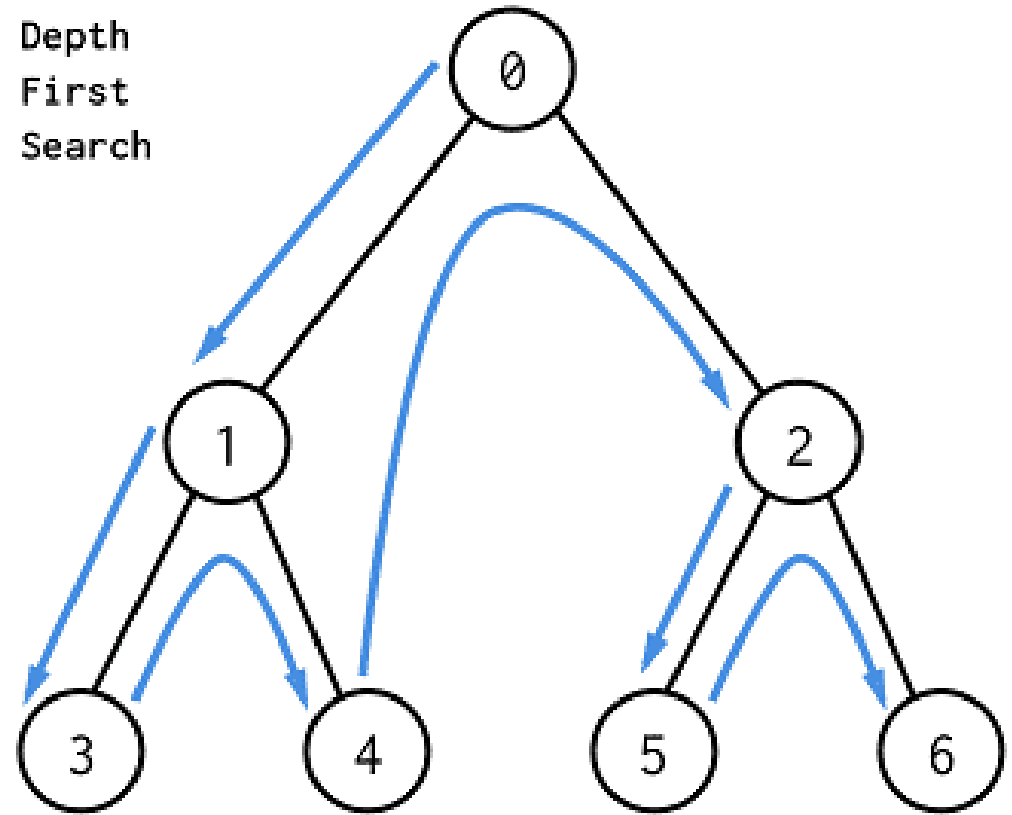


Search Algo Examples

Breadth
First
Search



Depth
First
Search



BFS	DFS
Queue based	Stack based
FIFO	LIFO

Pseudocode BFS

```
BFS (G, s)           //Where G is the graph and s is the source node
    let Q be queue.
    Q.enqueue( s )
    //Inserting s in queue until all its neighbour vertices are marked.

    mark s as visited.
    while ( Q is not empty)
    //Removing that vertex from queue,whose neighbour will be visited now
        v = Q.dequeue( )

        //processing all the neighbours of v
        for all neighbours w of v in Graph G
            if w is not visited
                Q.enqueue( w )
            //Stores w in Q to further visit its neighbour
            mark w as visited.
```

Pseudocode DFS

```
DFS (G, s):           //Where G is graph and s is source vertex
    let S be stack
    S.push( s )        //Inserting s in stack
    mark s as visited.

    while ( S is not empty):
        //Pop a vertex from stack to visit next
        v = S.top( )
        S.pop( )
        //Push all the neighbours of v in stack that are not visited
        for all neighbours w of v in Graph G:
            if w is not visited :
                S.push( w )
                mark w as visited
```

Exercise

- Implement **Breadth-First Search (BFS)** and **Depth-First Search (DFS)** algorithms that operate on a **graph represented as a class**, and interact with **Queue** and **Stack** class objects, respectively.